

New quantum invariants and knot fiberedness

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joint work with P. Orland, T. Saunders-A'Court, J. Svoboda

arXiv:2512.23700

Stanford Topology Seminar

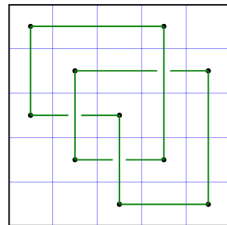
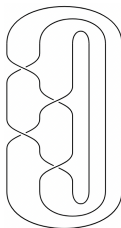
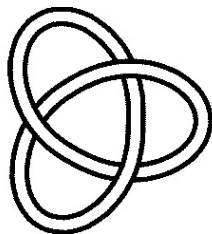
April 7, 2026

Motivation

Quantum invariants of knots

Let $K \subset S^3$ be a knot (embedding of a circle in the 3-sphere). Represent K in the plane by a knot diagram D .

Commonly used knot diagrams:

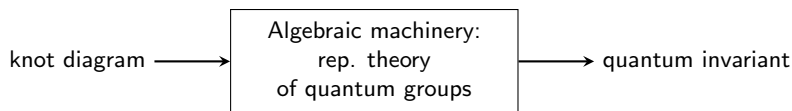


Key in the study of low-dimensional topology.

Motivation

Quantum invariants of knots

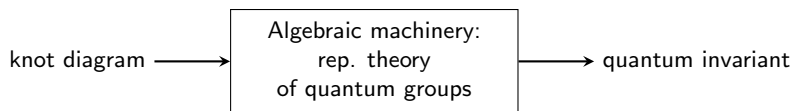
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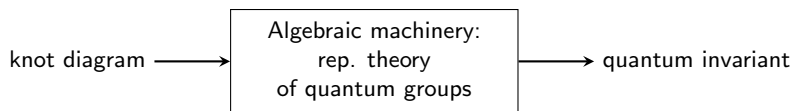
Why should we care about quantum topology?:

- 1 Strong distinguishing power.
- 2 Detects 4-dimensional information, incl. smooth vs. topological structures.

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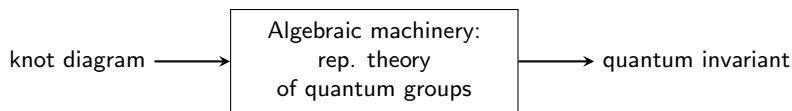
Question: what geometric content is read by quantum knot invariants?

Striking conjectures! $\rightsquigarrow$ volume, incompressible surfaces, genus...

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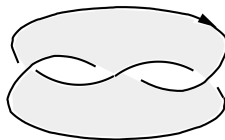
Our work: the $\mathfrak{sl}_2$ semisimple quantum knot invariants read existence of fibration over the knot complement.

Knots and fiberedness

Knots and surfaces

Knot $K \subset S^3 \rightsquigarrow$ connected, cpt, oriented surface S smoothly embedded in S^3 , $\partial S = K$. We call such a surface a Seifert surface of K .

Minimum genus of such surfaces $\rightsquigarrow g(K)$.



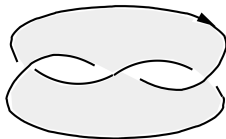
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Figure: N. Saveliev.

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What can we use Seifert surfaces for? We can construct the *Alexander polynomial* of a knot K .

Example: figure-eight

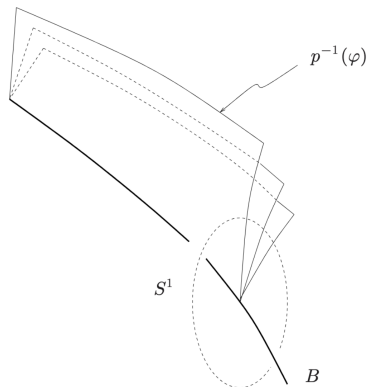
$$\Delta_{4_1}(x) = x^{-1} - 3 + x$$

Figure: N. Saveliev.

Fibered knots

Definition: A knot $K \subset S^3$ is **fibered** if there exists a fibration $S^3 \setminus K \rightarrow S^1$ with fiber F , $\partial F = K$.

We call the closure of F the (unique) fibered surface of K .



Equivalently, fibered knots are the binding of open book decompositions of S^3 .

Figure: H. Geiges.

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Proposition: $g(F) = g(K)$.

The Alexander polynomial is the characteristic polynomial of the map induced by the monodromy $h : F \rightarrow F$ on the first homology.

Theorem: Alexander polynomial of a fibered knot is monic and has degree $g(K)$.

$$\Delta_{4_1}(x) = x^{-1} - 3 + x$$

Hopf invariant or enhanced Milnor number

Let $N(K) \simeq D^2 \times K$ be a neighbourhood of K . We define a vector field ξ_K in S^3 :

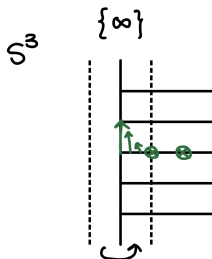
- $S^3 \setminus N(K)$: plane field tangent to the fibers.
- K : field orthogonal to the direction of K .
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Example: unknot.



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The **enhanced Milnor** number $\lambda(K)$ is the homotopy class of the plane field ξ_K .

$$\leadsto \lambda(K) = [\xi_K] \in \pi_3(S^2) \cong \mathbb{Z}.$$

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Equivalently, we can identify ξ_K with the contact structure induced by the open book decomposition of S^3 , which is bijection with $\mathbb{Z} \cup \{*\}$ (given by the Hopf map h).

$$\leadsto \lambda(K) = -h(\xi) \in \mathbb{Z}.$$

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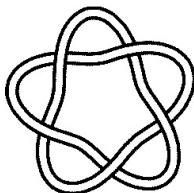
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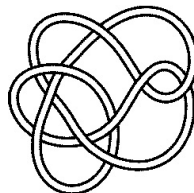
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Example: the knots 5_1 and 10_{132} have the same genus and Alexander polynomial, but **different Hopf invariant!**

$$\Delta_{5_1}(x) = \Delta_{10_{132}}(x) = x^{-2} - x^{-1} + 1 - x + x^2$$



$$\lambda(5_1) = 0$$



$$\lambda(10_{132}) = 2$$

Knot Floer homology

Knot Floer homology (Ozváth–Szabó, Rasmussen):

$$\widehat{HFK}(K) = \bigoplus_{n,s \in \mathbb{Z}} \widehat{HFK}_s(K, n).$$

Theorem: The graded Euler characteristic of $\widehat{HFK}$ is the Alexander polynomial $\Delta_K(x)$.

We say that knot Floer homology is a **categorification** of the Alexander polynomial.

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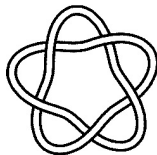
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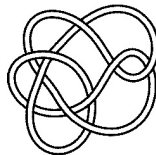
Examples:

$$\pi \left(\widehat{HFK}(5_1) \right) = m^{-4}x^{-2} + m^{-3}x^{-1} + m^{-2} + m^{-1}x + x^2$$

$$\pi \left(\widehat{HFK}(10_{132}) \right) = m^{-2}x^{-2} + (2m^{-1} + 1)x^{-1} + (2 + m) + (2m + m^2)x + m^2x^2$$



5₁



10₁₃₂

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Theorem: $\widehat{HFK}$ detects the genus: $\max \{n \mid \widehat{HFK}(K, n) \neq 0\} = g$.

Theorem (Ghiggini, Yi): $\widehat{HFK}$ detects fiberedness:

$$K \text{ is fibered} \iff \widehat{HFK}_s(K, g) = \begin{cases} \mathbb{Z} & s = \lambda(K) \\ 0 & \text{else} \end{cases} \quad \leftarrow \text{Hopf invariant}$$

We want to study whether quantum invariants detect fiberedness...

... But why do we care about quantum knot invariants in the first place?

Polynomials: Conjectured to be related to the knot volume, slopes of incompressible surfaces in the knot complement...

$$K \rightsquigarrow \{J_n(q)\}_{n \in \mathbb{Z}_{\geq 0}} \quad \leftarrow \text{colored Jones polynomials}$$

(Jones, Witten, Reshetikhin–Turaev, Garoufalidis–Le)

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Power series: The limit $q \nearrow 1$ of the sequence $\{J_n(q)\}_{n \in \mathbb{Z} \geq 0}$ agrees with

$$\frac{1}{\Delta_K(x)} + (q-1) \frac{P_K^{(1)}(x)}{\Delta_K^3(x)} + \cdots \in \mathbb{Q}[[x, q-1]]$$

where $x = q^n$. (Melvin–Morton, Rozansky, Bar Natan–Garoufalidis)

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It can be resumed as a 2-variable power series

$$F_K(x, q) = \sum_{i \in \frac{1}{2} + \mathbb{Z} \geq 0} f_i(q) x^i$$

for a certain class of knots, conjecturally all (Gukov–Manolescu, Park).

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Homology theories: $J_n(q)$ admits a **categorification** (Khovanov, Khovanov–Rozansky).

$F_K(x, q)$ is conjectured to admit a categorification as well, which would answer one of the central questions in quantum topology (Gukov–Pei–Putrov–Vafa).

Quantum knot invariants associated with $U_q(\mathfrak{sl}_2)$

State sum model

The quantum group $U_q(\mathfrak{sl}_2)$ is a “ q -deformation” of $\mathfrak{sl}_2$. Generated by $\{E, F, K, H\}$ and the set of relations:

$$[H, E] = 2E, \quad [H, F] = 2F, \quad [E, F] = \frac{K - K^{-1}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}}, \quad K^{\pm \frac{1}{2}} = q^{\pm \frac{H}{4}}$$

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and braiding

$$R = q^{\frac{H \otimes H}{4}} \sum_{k \geq 0} \frac{(q^{\frac{1}{2}} - q^{-\frac{1}{2}})^k}{[k]_q} q^{k(k-1)/4} E^k \otimes F^k \in U_q(\mathfrak{sl}_2) \hat{\otimes} U_q(\mathfrak{sl}_2)$$

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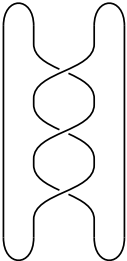
where $[k]_q := \frac{q^{\frac{k}{2}} - q^{-\frac{k}{2}}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}}.$

The action of R on the tensor product of $U_q(\mathfrak{sl}_2)$ representations induces a representation of the braid group.

Example of the state sum: trefoil

For generic q : $\{\text{simple f.d. representation } V_n\} \leftrightarrow \{\text{dim. of the vector space } (n+1)\}$.

Example: trefoil. The semisimple invariants (colored Jones) are computed via:


$$\xrightarrow{J_n} \text{tr}_{V_n \otimes V_n}^q \begin{pmatrix} V_n \otimes V_n \\ \uparrow R \\ V_n \otimes V_n \\ \uparrow R \\ V_n \otimes V_n \\ \uparrow R \\ V_n \otimes V_n \end{pmatrix} = \begin{cases} q + q^3 - q^4 & n = 1 \\ q^2 + q^5 - q^7 + q^8 - q^9 - q^{10} + q^{11} & n = 2 \\ \dots & \end{cases}$$

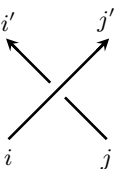
Initial construction of the F_K series

Let $V_{q^{-1}x}^h$ be the infinite-dimensional representation with highest weight $q^{-1}x$.

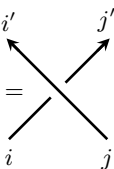
The R -matrix acting on $V_{q^{-1}x}^h \otimes V_{q^{-1}x}^h$ is:

$$R^{\pm 1} \cdot (v_i \otimes v_j) = \sum_{i+j=i'+j'} (R^{\pm 1})_{i,j}^{i',j'} (v_{i'} \otimes v_{j'})$$

In pictures:

$$R_{i,j}^{i',j'} =$$


Positive crossing

$$(R^{-1})_{i,j}^{i',j'} =$$


Negative crossing

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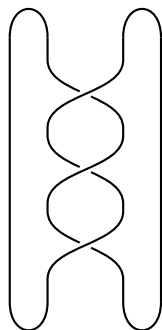
where

$$R_{i,j}^{i',j'} = (\text{polynomial in } q) x^{\frac{j+j'+1}{2}} (q^{j+1}x; q)_{i-j'} \in \mathbb{Z}[q][x]$$

$$\text{and } (R^{-1})_{i,j}^{i',j'} = R_{j,i}^{j',i'} \Big|_{\substack{q \rightarrow q^{-1} \\ x \rightarrow x^{-1}}} \in \mathbb{Z}[q^{-1}][x^{-1}].$$

$$(a; q)_n := (1-a)(1-aq) \cdots (1-aq^{n-1}).$$

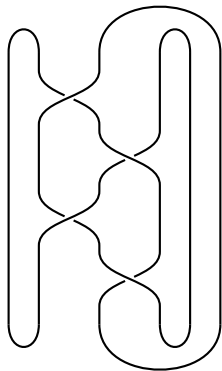
Trefoil knot



$$\xrightarrow{F_K} \operatorname{tr}_{V \otimes V}^q \begin{pmatrix} V \otimes V \\ \uparrow R \\ V \otimes V \\ \uparrow R \\ V \otimes V \\ \uparrow R \\ V \otimes V \end{pmatrix} = \sum_{i \geq 0} R_{0,i}^{i,0} R_{i,0}^{i,0} R_{i,0}^{0,i}$$

$$= qx - q^2 x^3 - q^3 x^4 + q^6 x^6 + q^8 x^7 - q^{13} x^9 - q^{16} x^{10} + q^{23} x^{12} + q^{27} x^{13} - q^{36} x^{15} + \dots$$

Figure eight knot



$$\xrightarrow{F_K} \operatorname{tr}_{V^{\otimes 3}}^q(\beta) = \sum_{i,j \geq 0} (R^{-1})_{i,j}^{i,j} R_{0,i}^{i,0} (R^{-1})_{0,j}^{0,j} R_{i,0}^{0,i}$$

$$= \sum_{i,j \geq 0} g_{i,j}(x, q) x^{i-2j}, \text{ where } g_{i,j}(x, q) \in \mathbb{Z}[q, q^{-1}, x].$$

Extension to fibered knots

Main issue: We need to modify the R -matrices to ensure the quantum trace gives us **only positive powers** of x .

$$R_{i,j}^{i',j'} = (\text{polynomial in } q) x^{\frac{j+j'+1}{2}} (q^{j+1}x; q)_{i-j'}$$
$$(R^{-1})_{i,j}^{i',j'} = (\text{polynomial in } q^{-1}) x^{-\frac{j+i'+1}{2}} (q^{-i-1}x^{-1}; q^{-1})_{j-i'}$$

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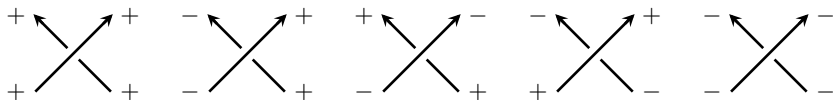
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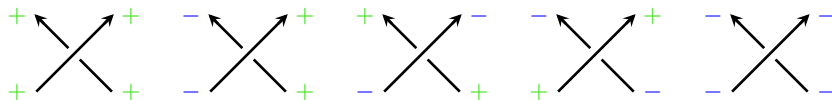
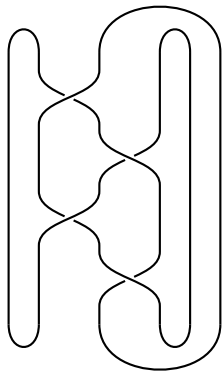


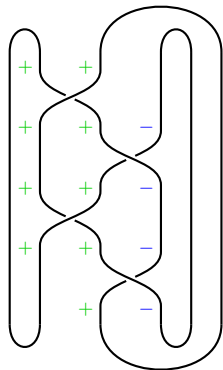
Figure eight knot



$$\xrightarrow{F_K} \operatorname{tr}_{V^{\otimes 3}}^q(\beta) = \sum_{i,j \geq 0} (R^{-1})_{i,j}^{i,j} R_{0,i}^{i,0} (R^{-1})_{0,j}^{0,j} R_{i,0}^{0,i}$$

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Figure eight knot (after the extension)



$$\xrightarrow{F_K} \mathrm{tr}_{V \otimes V}^q(\beta) = \sum_{i \geq 0, j < 0} (R^{-1})_{i,j}^{i,j} R_{0,i}^{i,0} (R^{-1})_{0,j}^{0,j} R_{i,0}^{0,i}$$

$$= 1 + 2x + (q^{-1} + 3 + q)x^2 + (2q^{-2} + 2q^{-1} + 5 + 2q + 2q^2)x^3 + \dots$$

The class of nice knots

Definition: A knot K is *nice* if there exists a braid representative of K admitting a $\{\pm\}$ -coloring of its segments such that $\mathrm{tr}^q(\beta) \in \mathbb{Z}[q, q^{-1}][[x]]$.

In that case, $F_K(x, q) := \mathrm{tr}^q(\beta)$.

The class of nice knots

Definition: A knot K is *nice* if there exists a braid representative of K admitting a $\{\pm\}$ -coloring of its segments such that $\mathrm{tr}^q(\beta) \in \mathbb{Z}[q, q^{-1}][[x]]$.

In that case, $F_K(x, q) := \mathrm{tr}^q(\beta)$.

Theorem (Park): Let K be a nice knot. Then, its F_K recovers the colored Jones polynomials perturbative expansion at the limit $q \nearrow 1$. In particular,

$$F_K(x, 1) \stackrel{x \sim 0}{=} \frac{1}{\Delta_K(x)} \in \mathbb{Z}[[x]].$$

The class of nice knots

Definition: A knot K is *nice* if there exists a braid representative of K admitting a $\{\pm\}$ -coloring of its segments such that $\mathrm{tr}^q(\beta) \in \mathbb{Z}[q, q^{-1}][[x]]$.

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Theorem (Park): Let K be a nice knot. Then, its F_K recovers the colored Jones polynomials perturbative expansion at the limit $q \nearrow 1$. In particular,

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Motivation: $\Delta_K(x)$ is **monic** whenever K is a fibered knot. Therefore,

$$\frac{1}{\Delta_K(x)} \in \mathbb{Z}[[x]] \subsetneq \mathbb{Q}[[x]].$$

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Theorem (Orland–S.–Saunder–A‘Court–Svoboda)

All fibered knots up to 12 crossings are nice.

The F_K series and fiberedness

What do these series have in common?

(Besides coming from fibered knots)

$$\begin{aligned}F_{4_1}(x, q) &= x + 2x^2 + (q^{-1} + 3 + q)x^3 \\&\quad + (2q^{-2} + 2q^{-1} + 5 + 2q + 2q^2)x^4 \\&\quad + (q^{-4} + 3q^{-3} + 4q^{-2} + 5q^{-1} + 8 + 5q + 4q^2 + 3q^3 + q^4)x^5 + \dots\end{aligned}$$

$$\begin{aligned}F_{8_{20}}(x, q) &= x^2 + x^3 + (1 - q - q^3)x^4 \\&\quad + (-q - q^3 - q^4)x^5 + (-q^{-1} - q - q^3 - q^4 + q^6 + q^9)x^6 \\&\quad + (-q^{-2} - q^{-1} - q^3 - q^4 + q^5 + q^6 + q^7 + q^9 + q^{10} + q^{11})x^7 + \dots\end{aligned}$$

$$\begin{aligned}F_{10_{153}}(x, q) &= q^{-1}x^3 + (-q^{-3} + q^{-1})x^4 + (q^{-6} - q^{-3} + q^{-1})x^5 \\&\quad + (-q^{-10} + q^{-6} - 2q^{-3})x^6 \\&\quad + (q^{-15} - q^{-10} + q^{-7} + q^{-6} + q^{-5} - 2q^{-3} - 1)x^7 + \dots\end{aligned}$$

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Fiberedness and Quantum Knot Invariants

Let K be a fibered knot. Denote by $g(K)$ its genus and by $\lambda(K)$ the Hopf invariant associated to the fiber surface of K .

Theorem (Orland–S.–Saunders–A'Court–Svoboda)

Let K be a nice knot. Then, F_K satisfies the following:

- 1 The leading term in x is of the form $q^{\ell(K)}$ for some $\ell(K)$.
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Conjecture (Orland–S.–Saunder–A'Court–Svoboda)

$$\{\text{fibered knots}\} = \{\text{nice knots}\}.$$

From sign diagrams to the Hopf invariant

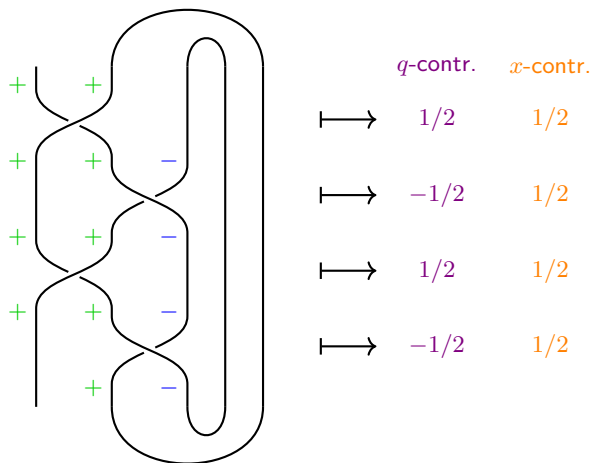
Sign diagrams contain **information about the fibered surface** of the knot!

Given a braid and a sign diagram, assign to each of its crossings:

	sign patterns	q -contr.	x -contr.
Positive crossing	$\begin{matrix} ++ & , & -+ \\ ++ & , & -+ \\ +- & , & -+ \\ -+ & , & +- \end{matrix}$	$+1/2$	$+1/2$
	$\begin{matrix} +- & , & -+ \\ -+ & , & +- \end{matrix}$	0	0
	$\begin{matrix} -- \\ -- \end{matrix}$	$+1/2$	$-1/2$
Negative crossing	$\begin{matrix} -- & , & +- \\ -- & , & +- \\ +- & , & -+ \\ -+ & , & +- \end{matrix}$	$-1/2$	$+1/2$
	$\begin{matrix} +- & , & -+ \\ -+ & , & +- \end{matrix}$	0	0
	$\begin{matrix} ++ \\ ++ \end{matrix}$	$-1/2$	$-1/2$

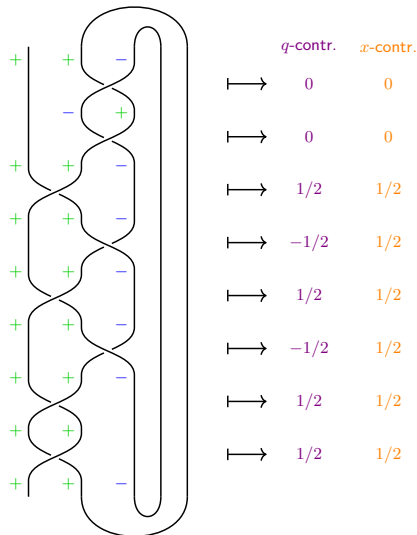
The sum of the q -contributions and x -contributions (with some extra braid-dependent factors) give $g(K) - \lambda(K)$ and $g(K)$ respectively.

Example: figure-eight



$$q\text{-deg} = \underbrace{0}_{\text{shift}} + 0 = 0 = g(4_1) - \lambda(4_1), \quad x\text{-deg} = \underbrace{-1}_{\text{shift}} + 2 = 1 = g(4_1)$$

Example: 8_{21}



$$q\text{-deg} = \underbrace{0}_{\text{shift}} + 1 = 1 = g(8_{21}) - \lambda(8_{21}), \quad x\text{-deg} = \underbrace{-1}_{\text{shift}} + 3 = 2 = g(8_{21})$$

Beyond fibered knots

Beyond fibered knots

- For fibered knots:

- ▶ $F_K(x, 1) \stackrel{x \sim 0}{=} \frac{1}{\Delta_K(x)} \in \mathbb{Z}[[x]]$
- ▶ F_K has finite q -polynomials as x -coefficients.

- Now, for **non-fibered knots**:

- ▶ $F_K(x, 1) \stackrel{x \sim 0}{=} \frac{1}{\Delta_K(x)} \in \mathbb{Q}[[x]]$
- ▶ In general, F_K will have **infinite q -series** as its x -coefficients!

There is a large class of knots for which we are able to make sense of the (now ill-defined) inverted state sum, including **strongly quasipositive knots**.

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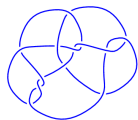
There is a large class of knots for which we are able to make sense of the (now ill-defined) inverted state sum, including **strongly quasipositive knots**.

Example:

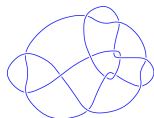
$$\begin{aligned} F_{9_{49}}(x, q) = & q^{-2} (-1 + 2q - q^2 - 3q^3 + 2q^4 + 2q^5 + 3q^6 - 3q^7 - 4q^8 + \dots) x^2 \\ & + q^{-2} (-2 + 4q + q^2 - 8q^3 - 2q^4 + 4q^5 + 13q^6 + 4q^7 - 8q^8 + \dots) x^3 \\ & + q^{-3} (1 - 5q + 7q^2 + 7q^3 - 11q^4 - 14q^5 - 8q^6 + 25q^7 + \dots) x^4 + O(x^5). \end{aligned}$$

What happens when a non-fibered knot has monic Alexander?

Two very different knots:



- Fibered, nice knot.
- $\widehat{HFK}(K, g) = x^3$
- $F_K = q^{-3}x^3 - 2q^{-3}x^4 + (q^{-5} + 4q^{-4} + 2q^{-2})x^5 + O(x^6)$



- Non-fibered, not-nice knot.
- $\widehat{HFK}(K, g) = (2 + m)x^3$
- $F_K = q^{-3}(1 - q^2 + 2q^4 - q^6 - q^7 + 2q^9 - q^{10} - q^{12} + \dots)x^3 + q^{-2}(1 - 3q - 2q^2 + 4q^3 + 2q^4 - 2q^5 - 3q^6 + 5q^8 + \dots)x^4 + q^{-4}(1 - 3q + 5q^2 - 3q^3 - 5q^4 + 7q^5 + 5q^6 - q^7 + \dots)x^5 + O(x^6)$

Same Alexander polynomial:

$$\Delta_K(x) = \frac{1}{x^3} + \frac{1}{x^2} - \frac{5}{x} + 7 - 5x + x^2 + x^3$$

and same genus.

Figures: knotinfo.org.

$\ell(K)$ in knot Floer homology

Define $F_K(x, q) = q^{\ell(K)} x^{d(K)} +$ (higher order terms in q or x).

Conjecture: $d(K) = -\ell(K) = g(K)$.

Example:

$$F_{9_{49}}(x, q) = q^{-2} (-1 + 2q - q^2 - 3q^3 + 2q^4 + 2q^5 + 3q^6 - 3q^7 - 4q^8 + \dots) x^2 \\ + q^{-2} (-2 + 4q + q^2 - 8q^3 - 2q^4 + 4q^5 + 13q^6 + 4q^7 - 8q^8 + \dots) x^3 + O(x^4).$$

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Conjecture: Let K be a fibered or strongly quasinegative. Define

$$\mu(K) := \min \{s \mid \widehat{HFK}_s(K, g(K)) \neq 0\}$$

Then, $d_K = g(K)$ and $\ell(K) = \mu(K)$.

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The previous conjecture is based on:

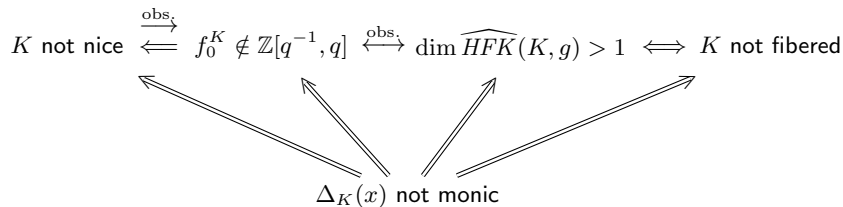
Observation: Let K be a strongly quasipositive knot of crossing number ≤ 16 . Then, $\mu(K) = 0$.

f_0^K seems to have greater distinguishing power than $\widehat{HFK}(K, g)$

$\pi(\widehat{HFK}(K, g))$	$\pm q^{-\ell(K)} f_0^K(q)$	K
1	1	fibred knots
2	$1 - q + q^3 - q^6 + q^{10} - q^{15} + q^{21} - \dots$	$5_2, 7_3, 7_5, 9_{3,6,9}, 10_{128}$
3	$1 - q + q^5 - q^8 + q^{16} - q^{21} + q^{33} + \dots$	$7_2, 9_{47}, 12n_{581}$
3	$1 - 2q + q^2 + 3q^3 - 2q^4 + \dots$	$8_{15}, 9_{49}, 10_{49,66,80}$
4	$1 - 2q + q^2 + 2q^3 - 2q^4 - q^6 + \dots$	$7_4, 9_{10,13,18,23}$
5	$1 - 3q + 3q^2 + 4q^3 - 8q^4 - 2q^5 + \dots$	9_{38}
5	$1 - 2q + q^2 + q^3 - q^4 + 2q^5 - 2q^6 + \dots$	$10_{55,63}$
6	$1 - 2q + q^2 + q^3 + \dots$	9_5
6	$1 - 3q + 3q^2 + 3q^3 - 7q^4 + \dots$	10_{53}
7	$1 - 2q + q^2 + 3q^5 - 3q^6 - q^7 - q^8 + \dots$	9_{35}
7	$1 - 3q + 3q^2 + 2q^3 - 6q^4 + 4q^5 + \dots$	10_{101}
8	$1 - 4q + 6q^2 + 3q^3 - 17q^4 + \dots$	10_{120}
$1 + 2m$	$1 + 2q + q^2 + q^3 + 2q^4 + 2q^5 + 2q^6 + \dots$	$12n_{801}$
$1 + m$	$1 + q + q^4 + q^9 + q^{16} + q^{25} + q^{36} + \dots$	$12n_{293,321,750,830}$

where $F_K(x, q) = f_0^K(q)x^{d_K} + O(x^{d_K+1})$.

We observe that $\widehat{HFK}(K_1, g_1) \cong \widehat{HFK}(K_2, g_2) \Rightarrow f_0^{K_1}(q) = f_0^{K_2}(q)$.



$$\begin{array}{c}
 \min_q f_0^K \xleftrightarrow{\text{obs.}} \min \{s \mid \dim \widehat{HFK}_s(K, g) \neq 0\} \\
 \\
 K \text{ not nice} \xleftrightarrow{\text{obs.}} f_0^K \notin \mathbb{Z}[q^{-1}, q] \xleftrightarrow{\text{obs.}} \dim \widehat{HFK}(K, g) > 1 \iff K \text{ not fibered} \\
 \\
 \swarrow \quad \nwarrow \quad \nearrow \quad \searrow \\
 \Delta_K(x) \text{ not monic}
 \end{array}$$

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Thank you!

Appendix slides

Hopf invariant or enhanced Milnor number

Alt. formal definition: Let ξ_1, ξ_2 be two no-where zero vector fields in general position in S^3 . Let

$$\Delta(\xi_1, \xi_2) := \{p \in S^3 \mid \xi_1(p) = t \xi_2(p) \text{ for some } t > 0\}.$$

Define the enhanced Milnor invariant or *Hopf invariant* of ξ_K as

$$\lambda(\xi_K) \equiv \lambda(K) := lk(\Delta(\xi_K, \xi_{std}), \Delta(\xi_K, -\xi_{std}))$$

where ξ_{std} is the canonical contact structure of S^3 and lk denotes the linking number.

Representation theory at generic q

$$[H, E] = 2E, \quad [H, F] = 2F, \quad [E, F] = \frac{K - K^{-1}}{q^{\frac{1}{2}} - q^{-\frac{1}{2}}}, \quad K^{\pm \frac{1}{2}} = q^{\pm \frac{H}{4}}$$

$$R = q^{\frac{H \otimes H}{4}} \sum_{k \geq 0} \frac{(q^{\frac{1}{2}} - q^{-\frac{1}{2}})^k}{[k]_q} q^{k(k-1)/4} E^k \otimes F^k \in U_q(\mathfrak{sl}_2) \hat{\otimes} U_q(\mathfrak{sl}_2)$$

Semisimple invariants

For generic $q \in \mathbb{C}$: for every $n \in \mathbb{N}$ there is a simple $U_q(\mathfrak{sl}_2)$ representation V_n of dimension $n + 1$.

$$E \cdot v_j = q^{\frac{n-2j}{4}} [j]_q v_{j-1}, \quad F \cdot v_j = q^{-\frac{n-2j-2}{4}} [n-j]_q v_{j+1}, \quad K \cdot v_j = q^{\frac{n-2j}{2}} v_j$$

$$j = 0, \dots, n.$$

.

New R -matrix formulas

extended R -matrix	condition
$R_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{jj'} (qx)^{\frac{j+j'+1}{2}} \left[\begin{matrix} i \\ i-j' \end{matrix} \right]_q (q^{j+1}x; q)_{i-j'}$	$i \geq j' \geq 0 \\ 0 > i \geq j'$
$R_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{jj'} (qx)^{\frac{j+j'+1}{2}} \left[\begin{matrix} i \\ j' \end{matrix} \right]_q \frac{1}{(q^jx; q^{-1})_{j'-i}}$	$j' \geq 0 > i$
$(R^{-1})_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{-ii'} (qx)^{-\frac{i+i'+1}{2}} \left[\begin{matrix} j \\ j-i' \end{matrix} \right]_{q^{-1}} (q^{-i-1}x^{-1}; q^{-1})_{j-i'}$	$j \geq i' \geq 0 \\ 0 > j \geq i'$
$(R^{-1})_{i,j}^{i',j'} = \delta_{i+j}^{i'+j'} q^{-ii'} (qx)^{-\frac{i+i'+1}{2}} \left[\begin{matrix} j \\ i' \end{matrix} \right]_{q^{-1}} \frac{1}{(q^{-i}x^{-1}; q)_{i'-j}}$	$i' \geq 0 > j$

Habiro series

Habiro coefficients $a_n(q) \in \mathbb{Z}[q^{\pm 1}]$, $n \geq 0$. Habiro series:

$$H_K(y, q) = \sum_{n=0}^{\infty} a_n(q) \sigma_n(y, q) \in \Lambda = \varprojlim_n \mathbb{Z}[q^{\pm 1}][y]/(\sigma_n) \quad (5.1)$$

where

$$H_K(q^n + q^{-n} - 2, q) = J_n(q) \quad \forall n \geq 0. \quad (5.2)$$

Remark: $\{J_n\} \leftrightarrow \{a_n\}$.

Theorem (Garoufalidis–Bar-Natan, Rozansky, Habiro): There exist polynomials $\{P_K^{(j)}(y)\}_{j \geq 0} \subseteq \mathbb{Z}[y]$ such that

$$H_K(y, 1 + h) = \frac{1}{\Delta_K(x)} + \sum_{i=1}^{\infty} \frac{P_K^{(j)}(x)}{\Delta_K(x)^{2j+1}} h^j \quad (5.3)$$

where the equality holds when expanding at $h = 0$ and $y = x + x^{-1} - 2 = 0$.